

Sabanci University

Spring 2025–2026

CS 201 – Programming Fundamentals

Course Information

Level of Course	Undergraduate
Prerequisites	IF 100
Course Page	https://sucourse.sabanciuniv.edu/plus/course/view.php?id=13251

Time & Place

Section A

Day	Time	Place
Monday	12:40-14:30	SBS-G071
Wednesday	12:40-13:30	FASS-G062

Section B

Day	Time	Place
Monday	08:40-10:30	FASS-G062
Wednesday	13:40-14:30	UC-G030

Instructor Information

Section A

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Section B

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Catalog Course Description

This course is intended to introduce students to the field of computing (basic computer organization, data representation, concepts, algorithmic thinking and problem solving), as well as give them intermediate level programming abilities in an object-oriented programming language

Course Learning Outcomes

1. Upon successful completion of Introduction to Computing, students are expected to be able to: Describe the basics of computer architecture, programming languages and compilers
2. Design an algorithm (step-by-step solution) for a given computing problem
3. Write small C++ programs
4. Use the basic programming concepts like if-else statements and while-for loops
5. Use functions and describe different parameter passing methods
6. Use, modify existing classes and design new classes
7. Perform simple text file I/O operations
8. Perform searches on arrays and sort arrays
9. Perform basic complexity analysis on algorithms

Course Objective

To introduce students to the field of computing (basic computer organization, data representation, concepts, algorithmic thinking and problem solving), as well as giving them intermediate level programming abilities in an object-oriented programming language (currently C++).

Course Materials

Resources

- All course materials will be available on SUCourse, including:
 - Lecture notes
 - Recitation samples
 - Example code

Textbook: A Computer Science Tapestry, 2nd Edition, by Owen L. Astrachan (not available in the bookstore anymore, but available at the library and online).

We may not follow the textbook at all times; you are responsible for all material covered in class.

Technology Requirements

- **Primary Recommendation:** Visual Studio (Community Edition, Windows)
- Visual Studio will serve as the main IDE for this course.
- It offers a full-featured environment for C++ development, including a built-in compiler, debugger, and project management tools.
- Students using Windows should install the latest version of Visual Studio Community Edition (free for educational use).
- **Alternative Option:** Visual Studio Code (Cross-Platform, Lightweight)
- VS Code may be used if you prefer a lighter IDE, or if you are working on macOS/Linux.
- It requires additional setup for compiling and debugging C++ programs.
- **Additional Setup for macOS Users (when using VS Code):**
 1. Install Xcode Command Line Tools by running: `xcode-select --install`
 2. Ensure a C/C++ compiler is available (usually provided with the Xcode tools).
 3. Install the C/C++ extension in VS Code (by Microsoft).
- Note: While both IDEs are acceptable, in-class demonstrations and examples will primarily use Visual Studio.

Assessment

Assessment Method	Percentage (%)
Final	35%
Midterm x 2	40%
Participation	6%
Attendance	4%
Homework	15%

Grading Clarification

- **Recitation Attendance – 4%:** Attendance in recitation sessions is mandatory. This component accounts for 4% of your total grade. Recitations are used for reinforcement, clarifying homework, and hands-on problem-solving.
- **Lecture Participation – 6%:** Class participation will be tracked and evaluated through TopHat activities during lectures. To receive credit, students must be present and participate in these activities. The quiz grade will be based 50% on participation and 50% on correctness.
- **Homework Assignments – 15%:** There will be 5 code-based homework assignments, distributed and submitted via SUCourse+. All homework assignments will be graded individually and may or may not have the same weight. Recitation sessions will be used for clarifying homework requirements and discussing solutions.
- **Midterms – 40%:** There will be a total of two midterms, each of which is weighted 20% of the overall grade, and the dates are to be announced later. The midterms will assess both your theoretical knowledge of the subjects studied in class and the practical knowledge gained through homework and recitations.
- **Final – 35%:** The final will account for 35% of the overall grade and will be scheduled by Student Resources. The final will assess both your theoretical knowledge of the subjects studied in class and the practical knowledge gained through homework and recitations.

Policies

Make-Up

- No makeups for participation and homework.
- If you miss one exam (either a midterm or the final) due to a documented valid excuse (e.g., a medical report), you will be allowed to take a make-up exam after the final exams. You may not miss more than one exam. If more than one exam is missed, the make-up exam will replace the highest-weighted missed exam, and the remaining missed exam(s) will receive a grade of zero.
- Make-up exams are often more comprehensive and cover all topics taught in the course from start to end.

Late Submission

Late submission will be accepted for up to two days, with a penalty of 10 points per late day. Submissions made within the first hour after the deadline will receive a reduced penalty of 5 points.

Academic Integrity

- Sharing work or submitting similar solutions (with other students or AI) is considered plagiarism.
- Plagiarism cases will be referred to the Dean’s Office for disciplinary action.

Special Notes

- Students are responsible for announcements made in lectures, SUCourse+, and email.
- Regularly check your Sabancı University email.
- Failure to follow announcements is not a valid excuse.

Use of AI

- Since this is an introductory course, we strongly recommend that you avoid using AI tools (such as ChatGPT) as much as possible and use them very responsibly.
- You may use AI tools to enhance your learning and improve your understanding of the course material; however, generating complete homework solutions using these tools is not permitted.
- Homework submissions found to rely excessively on LLM-generated content may receive a grade of zero and/or require the student to complete an oral examination.
- The use of LLMs or any AI tools during quizzes or exams, or to generate complete homework submissions, is strictly prohibited. Such use constitutes a violation of academic integrity and will result in disciplinary action if detected.

Weekly Content

1. Week 1: Introduction to Programming Languages, Data Representation (bits and bytes), basic programming structure and concepts: identifiers, literals, symbols, variables, screen input/output
2. Week 2: Basic data types (int/double/char/bool) and basic arithmetic operations with their precedence, first C++ program
3. Week 3: Conditional statements (if-else), nested else-if statements, logical operators (&&, ||, !)
4. Week 4: Functions with/out return values, function prototypes, parameter passing (pass by value and by reference)
5. Week 5: Intro to String class, Loops (while, for, do-while)
6. Week 6: Loops (while, for, do-while), char data type, string, and character processing
7. Week 7: File I/O, console stream cin, input and output file streams, string streams
8. Week 8: Structs, enum, vectors/arrays, and vector operations
9. Week 9: Vector operations: insert/delete to a vector
10. Week 10: Vector of structs, matrix
11. Week 11: Classes and objects: using existing classes such as Dice, RandGen, Date
12. Week 12: Classes and objects: using and modifying existing classes such as Dice, RandGen, Date
13. Week 13: Recursion
14. Week 14: Sequential/binary search, Sorting (selection and insertion sort), and introduction to algorithm complexity analysis